Android Recovery 量产问题排查指南

版本号: 1.0
发布日期: 2023.03.16

广州星翼电子科技有限公司zdyzlinux

版本历史

版本号	日期	制/修订人	内容描述
1.0	2023.03.16	AWA1633	初始版本文档

广州星翼电子科技有限公司zdyzlinux

目　录

1 前言 **1**
1.1 文档介绍 1
1.2 目标读者 1
1.3 常见的 recovery 问题 1

2 recovery 问题分析手段 **2**
2.1 recovery 问题调试手段 2
2.2 分析 recovery 问题的 log 2
2.2.1 data 分区异常 2
2.2.2 系统服务加载失败 3
2.2.3 恢复出厂设置 3

3 data 分区异常 **5**
3.1 data 分区挂载失败进入 recovery 流程分析 5
3.2 卡在开机动画不进系统-data 分区挂载失败 5
3.2.1 问题描述 5
3.2.2 问题初步分析 5
3.2.3 问题排查步骤 6
3.2.4 问题解决方案 7
3.3 开机不进系统-data 分区读写异常 8
3.3.1 问题描述 8
3.3.2 问题初步分析 8
3.3.3 问题排查步骤 9

4 系统服务加载失败问题 **10**
4.1 Android 的 RescueParty 机制 10
4.2 关机后不能按开机键开机-系统服务加载失败 10
4.2.1 问题描述 10
4.2.2 问题初步分析 10
4.2.3 问题排查步骤 10
4.2.4 问题解决方案 11

5 恢复出厂设置问题 **13**
5.1 点击恢复出厂设置闪退 13
5.1.1 问题描述 13
5.1.2 问题初步分析 13
5.1.3 问题排查步骤 14
5.1.4 问题解决方案 15
5.2 点击恢复出厂设置之后机器一直黑屏并且没有重启到 recovery 界面 15

版权所有 © 珠海全志科技股份有限公司。保留一切权利

5.2.1 问题描述 15
5.2.2 问题初步分析 15
5.2.3 问题排查步骤 18
5.2.4 问题解决方案 19
5.3 恢复出厂设置失败 20
5.3.1 问题描述 20
5.3.2 问题初步分析 20
5.3.3 问题排查步骤 20
5.3.3.1 问题解决方案 21
5.4 刷完谷歌 gsi 后，无法在设置界面恢复出厂设置 21
5.4.1 问题描述 21
5.4.2 问题初步分析 22
5.5 开机过程中卡在了电池图标界面 22
5.5.1 问题描述 22
5.5.2 问题初步分析 22
5.5.3 问题排查步骤 22
5.5.4 问题解决方案 23

6 OTA 问题 24
6.1 ota 差分包升级异常 24
6.1.1 问题描述 24
6.1.2 问题初步分析 24
6.1.3 问题排查步骤 25
6.1.4 问题解决方案 26

广州星翼电子科技有限公司zdyzlinux

1 前言

1.1 文档介绍

- 本文档提供 Android recovery 量产问题快速排查指导
- 本文档提供的快速排查指南工程思路如下：常规问题场景 + 日志分析 + 解决思路
- 本文档仅用于帮助开发人员快速排查和解决初级问题问题，如果按照文档中的指导仍无法解决问题，请联系全志原厂 FAE 进一步协助解决，并提供说明是否按照文档中的排查方案进行过必要的排查，提供排查过程的必要记录和数据

1.2 目标读者

本文档主要适用于以下工程师:

- 对 Android recovery 模块 “零” 基础的研发或 FAE 工程人员
- 对 Android recovery 模块初级基础的研发或 FAE 工程人员

1.3 常见的 recovery 问题

进入 Android recovery 的问题可以归为以下四类：

（1）data 分区异常

（2）系统服务加载失败

（3）恢复出厂设置问题

（4）OTA 问题

版权所有 © 珠海全志科技股份有限公司。保留一切权利

2 recovery 问题分析手段

2.1 recovery 问题调试手段

```
1. 打开串口log。
reboot uboot
printenv 查看cma的大小，如cma=8M。则设置cma为8M，并附带printk.devkmsg=on参数
setenv cma 8M printk.devkmsg=on
setenv loglevel 8
saveenv

2. 对于系统启动失败问题，在系统起来的时候，立刻在串口中输入logcat，抓取logcat。

3. 在复现recovery问题的时候，接串口进行测试。

4. 抓recovery系统的启动log：recovery模式中，log在/tmp/recovery.log文件中。

5. 排除ddr和flash物料的问题。
```

2.2 分析 recovery 问题的 log

2.2.1 data 分区异常

```
常见的data分区异常的log错误信息。

（1）vdc。
vdc: Command: cryptfs mountFstab /dev/block/by-name/userdata /data Failed: Status(-8, EX_SERVICE_SPECIFIC): '0: '

（2）credstore。
[  14.857566] DEBUG: pid: 318, tid: 318, name: credstore  >>> /system/bin/credstore <<<
[  14.863379] DEBUG: uid: 1076
[  14.872287] DEBUG: signal 6 (SIGABRT), code -1 (SI_QUEUE), fault addr --------
[  14.875670] DEBUG: Abort message: 'Check failed: chdir(data_dir.c_str()) != -1 chdir: /data/misc/credstore: No such file
      or directory'

（3）F2FS-fs。
[   9.064250][  T193] F2FS-fs (dm-5): invalid crc_offset: 322151910
[   9.071302][  T193] F2FS-fs (dm-5): invalid crc_offset: 4198638403
[   9.078255][  T193] F2FS-fs (dm-5): Failed to get valid F2FS checkpoint

（4）logcat中搜vold。
12-31 21:10:20.520  187  187 E vold   : No key found in /metadata/vold/metadata_encryption/key
12-31 21:10:21.545  187  187 I Checkpoint: cp_prepareCheckpoint called
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

```
12-31 21:10:21.661  219  219 I derive_sdk: Read version 0 from /apex/com.android.sdkext/etc/sdkinfo.binarypb
```

2.2.2 系统服务加载失败

（1）查看crash信息：logcat -b crash。

（2）查看runtime的crash信息：logcat | grep AndroidRuntime。

（3）查看watchdog的信息：logcat | grep watchdog -i。

（4）查看system_server是否被杀掉：ps -Af | grep system_server，如果system_server的pid号很大，则system_server有重启过。然后logcat看system_server的log。

（5）查看zygote是否被杀掉：ps -Af | grep zygote，如果zygote的pid号很大，则zygote有重启过。然后logcat看zygote的log。

（6）抓取/data/anr的信息。

（7）抓取/data/tombstones的信息。

（8）查看system_server或者其他服务的堆栈信息：debuggerd -b 进程的pid号。

2.2.3 恢复出厂设置

（1）uncrypt的打印。

```
[ 384.480645][    T1] init: starting service 'setup-bcb'...
[ 384.501093][    T1] init: Created socket '/dev/socket/uncrypt', mode 600, user 1000, group 1000
[ 384.522933][    T1] init: Control message: Processed ctl.start for 'setup-bcb' from pid: 465 (system_server)
[ 385.599536][    T1] init: Service 'setup-bcb' (pid 5563) exited with status 0 oneshot service took 1.084000 seconds in
    background
[ 385.619476][    T1] init: Sending signal 9 to service 'setup-bcb' (pid 5563) process group...
```

（2）uboot关于misc分区的打印。

```
[01.872]update part info
bbe6bd08: 746f6f62 6365722d 7265766f 00000079    boot-recovery...
bbe6bd18: 00000000 00000000 00000000 00000000    ................
bbe6bd28: 00000000 00000000 00000000 00000000    ................
bbe6bd38: 00000000 00000000 00000000 00000000    ................
bbe6bd48: 00000000 00000000 00000000 00000000    ................
bbe6bd58: 00000000 00000000 00000000 00000000    ................
bbe6bd68: 00000000 00000000 00000000 00000000    ................
bbe6bd78: 00000000 00000000 00000000 00000000    ................
bbe6bd88: 00000000 00000000 00000000 00000000    ................
bbe6bd98: 00000000 00000000 00000000 00000000    ................
bbe6bda8: 00000000 00000000 00000000 00000000    ................
bbe6bdb8: 00000000 00000000 00000000 00000000    ................
bbe6bdc8: 00000000 00000000 00000000 00000000    ................
bbe6bdd8: 00000000 00000000 00000000 00000000    ................
bbe6bde8: 00000000 00000000 00000000 00000000    ................
bbe6bdf8: 00000000 00000000 00000000 00000000    ................
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

```
bbe6be08: 00000000 00000000 00000000 00000000    ................
bbe6be18: 00000000 00000000 00000000 00000000    ................
bbe6be28: 00000000 00000000 00000000 00000000    ................
bbe6be38: 00000000 00000000 00000000 00000000    ................
[02.012]bootmode[2]:0x5c
```

（3）Android 12及以前 cmdline的打印。

进入recovery的标志：androidboot.force_normal_boot=0

进入charger充电模式的标志：androidboot.mode=charger bootreason=charger

广州星翼电子科技有限公司zdyzlinux

3 data 分区异常

3.1 data 分区挂载失败进入 recovery 流程分析

从代码的角度分析，以下场景可能会进入到recovery。

1. rc文件，命令执行失败。

（1）perform_apex_config和mkdir，在设置文件夹的文件加密属性的时候报错的话，会进入到recovery。

（2）vdc命令，执行vdc命令失败的时候，会进入到recovery。

（3）mount_all命令，挂载data分区失败的时候，可能会进入到recovery。

（4）remount_userdata命令，remount data分区失败的时候，可能会进入到recovery。

3.2 卡在开机动画不进系统-data 分区挂载失败

3.2.1 问题描述

1. 出现卡在动画不进系统的机器，强制关机在开机也不进系统。

2. 出现卡在动画不进系统的机器，重新刷机后可以进系统，并且不容易再次复现。

3. 复现问题抓取的log见附件，初步查看mount data分区的时候出错。

3.2.2 问题初步分析

从串口log看：是data分区挂载失败，导致卡在动画不进系统。一般data分区挂载失败的比较明显的log标志就是credstore的crash log。

```
[   8.073005] vdc: Command: cryptfs mountFstab /dev/block/by-name/userdata /data Failed: Status(-8,
        EX_SERVICE_SPECIFIC): '0: '

[  14.829121] DEBUG: *** *** *** *** *** *** *** *** *** *** *** *** *** *** *** ***
[  14.829322] DEBUG: Build fingerprint: 'Multilaser/M7_WIFI/ML_JI1D_M7_WIFI:11/V2_20220112/2:userdebug/test-keys'
[  14.838146] DEBUG: Revision: '0'
[  14.849526] DEBUG: ABI: 'arm'
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

```
[  14.854128] DEBUG: Timestamp: 1969-12-31 21:00:35-0300
[  14.857566] DEBUG: pid: 318, tid: 318, name: credstore  >>> /system/bin/credstore <<<
[  14.863379] DEBUG: uid: 1076
[  14.872287] DEBUG: signal 6 (SIGABRT), code -1 (SI_QUEUE), fault addr --------
[  14.875670] DEBUG: Abort message: 'Check failed: chdir(data_dir.c_str()) != -1 chdir: /data/misc/credstore: No such file
      or directory'
[  14.883858] DEBUG:    r0  00000000  r1  0000013e  r2  00000006  r3  ffa98420
[  14.897467] DEBUG:    r4  ffa98434  r5  ffa98418  r6  0000013e  r7  0000016b
[  14.905670] DEBUG:    r8  ffa98420  r9  ffa98430  r10 ffa98450  r11 ffa98440
[  14.913742] DEBUG:    ip  0000013e  sp  ffa983f0  lr  ea0906a1  pc  ea0906b4
[  14.932862] DEBUG:
[  14.943189] DEBUG: backtrace:
[  14.943403] DEBUG:      #00 pc 000626b4  /apex/com.android.runtime/lib/bionic/libc.so (abort+172) (BuildId: 6103
      f0e8daf55e5cef6ad85f5e8bb4f8)
[  14.946915] DEBUG:      #01 pc 0000c4cb  /system/lib/libbase.so (android::base::DefaultAborter(char const*)+6) (BuildId
      : c02434d1b5d1ae1da4cf49a902da9601)
[  14.961547] DEBUG:      #02 pc 0000d991  /system/lib/libbase.so (android::base::SetAborter(std::__1::function<void (
      char const*)>&&)::$_3::__invoke(char const*)+48) (BuildId: c02434d1b5d1ae1da4cf49a902da9601)
[  14.978260] DEBUG:      #03 pc 0000d29d  /system/lib/libbase.so (android::base::LogMessage::~LogMessage()+224) (
      BuildId: c02434d1b5d1ae1da4cf49a902da9601)
[  14.999539] DEBUG:      #04 pc 00009ceb  /system/bin/credstore (main+622) (BuildId:
      c6aa0905300f804e1cac30f35b861bd9)
[  15.015510] DEBUG:      #05 pc 0005cec3  /apex/com.android.runtime/lib/bionic/libc.so (__libc_init+66) (BuildId: 6103
      f0e8daf55e5cef6ad85f5e8bb4f8)
```

3.2.3 问题排查步骤

```
1. 从串口log看，可以定位到是处理文件加密的时候出错了，导致挂载data分区失败。

vdc: Command: cryptfs mountFstab /dev/block/by-name/userdata /data Failed: Status(-8, EX_SERVICE_SPECIFIC): '0: '

2. 系统没挂，没进recovery，串口可用，可以抓取logcat。
从logcat看，处理文件加密的过程中，发现没有key文件，No key found in /metadata/vold/metadata_encryption/key，导
      致挂载data分区失败。
从串口进入到/metadata/vold/metadata_encryption/目录，也可以看到没有key文件。

12-31 21:10:20.519  187  187 I vold   : Using AllWiner-KeymasterDevice from AllWinner for encryption.  Security level:
      TRUSTED_ENVIRONMENT, HAL: android.hardware.keymaster@4.1::IKeymasterDevice/default
12-31 21:10:20.520  187  187 E vold   : No key found in /metadata/vold/metadata_encryption/key
12-31 21:10:21.545  187  187 I Checkpoint: cp_prepareCheckpoint called
12-31 21:10:21.661  219  219 I derive_sdk: Read version 0 from /apex/com.android.sdkext/etc/sdkinfo.binarypb
12-31 21:10:21.662  219  219 I derive_sdk: R extension version is 0
12-31 21:10:21.689  187  187 D vold   : fscrypt_init_user0
12-31 21:10:21.689  187  187 D vold   : fscrypt_prepare_user_storage for volume null, user 0, serial 0, flags 1
12-31 21:10:21.689  187  187 D vold   : Preparing: /data/system/users/0
12-31 21:10:21.689  187  187 E cutils : Failed to mkdir(/data/system/users/0): No such file or directory
12-31 21:10:21.689  187  187 E vold   : Failed to prepare /data/system/users/0: No such file or directory
12-31 21:10:21.689  187  187 E vold   : Failed to prepare user 0 storage

3. 排查key文件的生成过程
从代码中可以看到key文件的生成过程：
system/core/fs_mgr/fs_mgr.cpp和system/vold/MetadataCrypt.cpp文件
第一次烧固件起来的时候：
调用vdc cryptfs encryptFstab命令：
  (1) 生成key文件。
```

（2）格式化data分区。
（3）通过key以文件加密的方式挂载data分区。

从log看，vdc cryptfs mountFstab 命令，说明第一次烧固件起来的时候，已经对data分区格式化过了，说明key文件也已经生成过了。
所以可能是第一次烧固件起来的时候，机器异常掉电，key文件没有及时保存到flash中，然后第二次系统起来的时候，找不到key文件了，导致data分区挂载失败了。

3.2.4 问题解决方案

由于这个问题是机器异常掉电，key文件没有及时保存到flash导致的。
所以我们的解决方案就是要确保key文件真的写到flash中了。

1. 第一次烧固件起来的时候：
调用vdc cryptfs encryptFstab命令，如果这个命令执行成功，则往/metadata/vold/metadata_encryption/key/done文件写1。

2. 第二次系统起来的时候：
（1）如果/metadata/vold/metadata_encryption/key/done文件不存在，则格式化data分区，然后再走一次文件加密流程，就是调用vdc cryptfs encryptFstab命令。
（2）如果/metadata/vold/metadata_encryption/key/done文件存在，则说明不是key文件不存在的问题了，就走recovery，提示用户data分区异常，需要擦除data分区数据

补丁：

```
diff --git a/fs_mgr/fs_mgr.cpp b/fs_mgr/fs_mgr.cpp
index 9c4763254..ee9f471bd 100644
--- a/fs_mgr/fs_mgr.cpp
+++ b/fs_mgr/fs_mgr.cpp
@@ -90,6 +90,8 @@
 #define SYSFS_EXT4_VERITY "/sys/fs/ext4/features/verity"
 #define SYSFS_EXT4_CASEFOLD "/sys/fs/ext4/features/casefold"

+#define METADATA_DONE "/metadata/vold/metadata_encryption/key/done"
+
 // FIXME: this should be in system/extras
 #define EXT4_FEATURE_COMPAT_STABLE_INODES 0x0800

@@ -1478,6 +1480,9 @@ int fs_mgr_mount_all(Fstab* fstab, int mount_mode) {
                 return FS_MGR_MNTALL_FAIL;
             }
         }
+        if (!android::base::WriteStringToFile("1", METADATA_DONE)) {
+            LERROR << "not write done to /metadata/vold/metadata_encryption/key/done";
+        }
     }

     // Success!  Go get the next one.
@@ -1544,9 +1549,26 @@ int fs_mgr_mount_all(Fstab* fstab, int mount_mode) {
         encryptable = FS_MGR_MNTALL_DEV_MIGHT_BE_ENCRYPTED;
     } else if (mount_errno != EBUSY && mount_errno != EACCES &&
             should_use_metadata_encryption(attempted_entry)) {
+        if (access(METADATA_DONE, F_OK) == 0) {
+            LERROR << "/metadata/vold/metadata_encryption/key/done found";
+        }
         if (!call_vdc({"cryptfs", "mountFstab", attempted_entry.blk_device,
```

```
                attempted_entry.mount_point},
                nullptr)) {
+           if (access(METADATA_DONE, F_OK) != 0) {
+             if (fs_mgr_do_format(current_entry, false) == 0) {
+               // Let's replay the mount actions.
+               i = top_idx - 1;
+               continue;
+             } else {
+               LERROR << __FUNCTION__ << "(): Format failed. "
+                 << "Suggest recovery...";
+               encryptable = FS_MGR_MNTALL_DEV_NEEDS_RECOVERY;
+               continue;
+             }
+           } else {
+             LERROR << "/metadata/vold/metadata_encryption/key/done found in vdc";
+           }
          ++error_count;
        }
        encryptable = FS_MGR_MNTALL_DEV_IS_METADATA_ENCRYPTED;
```

3.3 开机不进系统-data 分区读写异常

3.3.1 问题描述

部分机器,开机不进系统

3.3.2 问题初步分析

1. 从串口log看，是f2fs文件系统异常，导致执行vdc命令失败，然后进recovery提示用户擦除分区数据

```
[   8.913030][  T193] fsck.f2fs: Info: Fix the reported corruption.
[   8.926663][  T193] fsck.f2fs:       Info: No support kernel version!
[   8.933426][  T193] fsck.f2fs: Info: Segments per section = 1
[   8.939933][  T193] fsck.f2fs: Info: Sections per zone = 1
[   8.946113][  T193] fsck.f2fs: Info: sector size = 4096
[   8.951988][  T193] fsck.f2fs: Info: total sectors = 3208059 (12531 MB)
[   8.959430][  T193] fsck.f2fs: Info: MKFS version
[   8.964719][  T193] fsck.f2fs:   "5.4.161-svn20439-ab20220707-152536"
[   8.971974][  T193] fsck.f2fs: Info: FSCK version
[   8.977255][  T193] fsck.f2fs:   from "5.4.161-svn20439-ab20220707-152536"
[   8.986811][  T193] F2FS-fs (dm-5): Using encoding defined by superblock: utf8-12.1.0 with flags 0x0
[   8.997455][  T193] F2FS-fs (dm-5): invalid crc_offset: 322151910
[   9.004526][  T193] F2FS-fs (dm-5): invalid crc_offset: 4198638403
[   9.011525][  T193] F2FS-fs (dm-5): Failed to get valid F2FS checkpoint
[   9.053568][  T193] F2FS-fs (dm-5): Using encoding defined by superblock: utf8-12.1.0 with flags 0x0
[   9.064250][  T193] F2FS-fs (dm-5): invalid crc_offset: 322151910
[   9.071302][  T193] F2FS-fs (dm-5): invalid crc_offset: 4198638403
[   9.078255][  T193] F2FS-fs (dm-5): Failed to get valid F2FS checkpoint
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

vdc: Command: cryptfs init_user0 Failed: Status(-8, EX_SERVICE_SPECIFIC): '0: '

2. 从串口log看，f2fs文件系统数据异常，导致系统服务多次启动失败，然后进入recovery。

```
[  41.548468][  T385] F2FS-fs (dm-5): inconsistent node block, nid:1886, node_footer[nid:3787,ino:3787,ofs:0,cpver
    :11985581956095338672,blkaddr:56292]
[  41.573815][  T385] F2FS-fs (dm-5): inconsistent node block, nid:3146, node_footer[nid:3335,ino:3335,ofs:0,cpver
    :11985581956095338672,blkaddr:56282]
[  41.595214][  T274] [LLL] DISP_LCD_SET_BRIGHTNESS: 0, mFlag=0, g_lcd_bl_force_disabled=0.
[  41.599485][  T385] F2FS-fs (dm-5): inconsistent node block, nid:3048, node_footer[nid:3184,ino:3184,ofs:0,cpver
    :11985581956095338672,blkaddr:56280]
```

3.3.3 问题排查步骤

对于出现f2fs文件系统异常的问题，需要排查ddr或者flash物料是否稳定。

广州星翼电子科技有限公司zdyzlinux

4 系统服务加载失败问题

4.1 Android 的 RescueParty 机制

安卓的自救：RescueParty
监控system_server出现循环崩溃的时候，RescueParty会根据不同的救援级别做出一系列操作，看是否可恢复设备，最严重的时候则是通过进入recovery，提示用户清空用户数据恢复出厂设置解决。

frameworks/base/services/java/com/android/server/SystemServer.java
PackageWatchdog.getInstance(mSystemContext).noteBoot()函数，每次启动system_server都会计数+1，然后将数据存储到/metadata/watchdog/mitigation_count.txt文件中，当重启到5次之后，就会进入到recovery，提示用户清除用户数据。

4.2 关机后不能按开机键开机-系统服务加载失败

4.2.1 问题描述

机器关机后不能按开机键开机,长时间导致电池耗尽

4.2.2 问题初步分析

从串口看，zygote一直挂，然后就进入recovery了，估计是system_server一直重启，重启够5次之后，触发RescueParty机制，然后重启进入recovery了。

4.2.3 问题排查步骤

1. 在串口可以输入的时候，立刻在串口输入logcat，抓取logcat，做进一步分析。
从logcat看，是awbms crash了，导致system_server一直重启，重启够5次之后，触发RescueParty机制，然后重启进入recovery。

```
[2022-08-26 10:55:28] --------- beginning of crash
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: *** FATAL EXCEPTION IN SYSTEM PROCESS: BackgroundManagerService
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: java.lang.NullPointerException: Attempt to invoke virtual method 'void com.android.server.wm.RootWindowContainer.getRunningTasks(int, java.util.List, int, int, android.util.ArraySet)' on a null object reference
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at com.android.server.wm.
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

```
    ActivityTaskManagerService.getTasks(ActivityTaskManagerService.java:2267)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at com.android.server.wm.
    ActivityTaskManagerService.getTasks(ActivityTaskManagerService.java:2237)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at com.android.server.am.
    ActivityManagerService.getTasks(ActivityManagerService.java:5882)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at com.softwinner.awbms.TaskController.
    doLoadCurrentTasks(TaskController.java:84)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at com.softwinner.awbms.TaskController.
    onTaskChange(TaskController.java:74)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at com.softwinner.awbms.TaskController.
    lambda$new$0$TaskController(TaskController.java:31)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at com.softwinner.awbms.
    TaskController$$ExternalSyntheticLambda0.run(Unknown Source:2)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at android.os.Handler.handleCallback(Handler.
    java:938)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at android.os.Handler.dispatchMessage(Handler.
    java:99)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at android.os.Looper.loopOnce(Looper.java:201)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at android.os.Looper.loop(Looper.java:288)
[2022-08-26 10:55:28] 08-24 09:30:10.245  454  510 E AndroidRuntime: at android.os.HandlerThread.run(HandlerThread
    .java:67)
```

4.2.4 问题解决方案

```
由于awbms服务启动的太早了，很多服务都还没初始化完整，导致获取某些服务信息的时候，触发了空指针crash。
解决办法：将awbms靠后启动。

---

diff --git a/services/java/com/android/server/SystemServer.java b/services/java/com/android/server/SystemServer.java
index 4913c82..8a0d4f3 100644
--- a/services/java/com/android/server/SystemServer.java
+++ b/services/java/com/android/server/SystemServer.java
@@ -1135,18 +1135,6 @@
     mIncrementalServiceHandle = startIncrementalService();
     t.traceEnd();

-    if (!SystemProperties.getBoolean("config.diable_awbms", false)) {
-      t.traceBegin("StartAwbms");
-      try {
-        new Thread(() -> {
-          startAwbms();
-        }).start();
-      } catch (Throwable e) {
-        reportWtf("starting Awbms", e);
-      }
-      t.traceEnd();
-    }
-
     // Power manager needs to be started early because other services need it.
     // Native daemons may be watching for it to be registered so it must be ready
     // to handle incoming binder calls immediately (including being able to verify
@@ -1654,6 +1642,18 @@
       throw e;
     }
```

```
+        if (!SystemProperties.getBoolean("config.diable_awbms", false)) {
+            t.traceBegin("StartAwbms");
+            try {
+                new Thread(() -> {
+                    startAwbms();
+                }).start();
+            } catch (Throwable e) {
+                reportWtf("starting Awbms", e);
+            }
+            t.traceEnd();
+        }
+
         // Before things start rolling, be sure we have decided whether
         // we are in safe mode.
         final boolean safeMode = wm.detectSafeMode();
```

广州星翼电子科技有限公司zdyzlinux

5 恢复出厂设置问题

5.1 点击恢复出厂设置闪退

5.1.1 问题描述

关机后不能按开机键开机,长时间导致电池耗尽。进入安卓之后，点“清除所有数据”闪退。

5.1.2 问题初步分析

```
1. 点击“清除所有数据”之后闪退，从代码流程以及log，我们分析到，是uncrypt服务起来了，导致闪退的。
packages/apps/Settings/src/com/android/settings/MainClearConfirm.java
onClick
  -> doMainClear
    -> 发送一个Intent.ACTION_FACTORY_RESET给MasterClearReceiver

frameworks/base/services/core/java/com/android/server/MasterClearReceiver.java
onReceive
  -> RecoverySystem.rebootWipeUserData

对应的log打印:
08-23 21:15:10.902  1000  2864  2864 W MasterClear: !!! FACTORY RESET !!!
08-23 21:15:10.903  1000  2864  2864 I MasterClear: NOT wiping external storage; starting thread Reboot
08-23 21:15:10.906  1000  2864  8696 I MasterClear: Calling RecoverySystem.rebootWipeUserData(context, shutdown=
      false, reason=MainClearConfirm, forceWipe=false, wipeEsims=false)

frameworks/base/core/java/android/os/RecoverySystem.java
rebootWipeUserData
  -> bootCommand
    -> rs.rebootRecoveryWithCommand(调用到RecoverySystemService服务)
      -> setupOrClearBcb
        -> checkAndWaitForUncryptService
          -> 检查init.svc.uncrypt或者init.svc.setup-bcb或者init.svc.clear-bcb属性是否为running，如果是running的话，说
      明有进程在使用他们，则会尝试30次，一次等1秒，一共等30s。最后还是得不到的话，就会闪退。

从bugreport中，我们可以看到这些属性是running的，所以需要查清楚，为啥uncrypt服务会起来。
[init.svc.setup-bcb]: [running]
[init.svc.uncrypt]: [running]
[init.svc.clear-bcb]: [running]

对应的log打印:
08-23 21:15:40.939  1000  2864  8696 E RecoverySystemService: uncrypt service is unavailable.
08-23 21:15:40.939  1000  2864  8703 W system_server: Long monitor contention with owner Reboot (8696) at void com.
      android.server.recoverysystem.RecoverySystemService.rebootRecoveryWithCommand(java.lang.String)(
      RecoverySystemService.java:516) waiters=0 in void com.android.server.recoverysystem.RecoverySystemService.
```

```
    rebootRecoveryWithCommand(java.lang.String) for 26.307s
08-23 21:15:40.939  1000  2864  8696 E MasterClear: Can't perform master clear/factory reset
08-23 21:15:40.939  1000  2864  8696 E MasterClear: java.io.IOException: Reboot failed (no permissions?)
08-23 21:15:40.939  1000  2864  8696 E MasterClear:    at android.os.RecoverySystem.bootCommand(RecoverySystem.
    java:1300)
08-23 21:15:40.939  1000  2864  8696 E MasterClear:    at android.os.RecoverySystem.rebootWipeUserData(
    RecoverySystem.java:998)
08-23 21:15:40.939  1000  2864  8696 E MasterClear:    at com.android.server.MasterClearReceiver$1.run(
    MasterClearReceiver.java:107)
```

5.1.3 问题排查步骤

```
分析所有复现的机器的logcat，都发现是一样的问题，uncrypt服务起来了，导致点“清除所有数据”闪退。
从串口log可以看到，所有的机器，都会有一个现象，就是休眠唤醒之后，时间戳变了。
从logcat看，zygote的pid是很大的，说明zygote重启过，但是没有zygote被杀掉的log，所以需要找到zygote重启或者安卓系
    统重启的原因。
最后从串口log中，看到了一句sysrq的打印，sysrq: Terminate All Tasks，说明是sysrq杀死了所有进程。
然后从uncrypt.rc文件中，可以看到uncrypt服务是class main类型的。
然后vndservicemanager服务被kill掉之后，重启的时候，会重新启动class main类型的服务，这时候，uncrypt服务就起来
    了，就导致点“清除所有数据”闪退。

08-23 20:18:45.569887[   84.689456] binder_alloc: allocated: 0 (num: 0 largest: 0), free: 1040384 (num: 1 largest: 1040384)
08-23 20:18:45.569928[   84.700506] binder: 1214:1489 transaction failed 29201/-28, size 1056768-0 line 2906
08-23 20:18:45.589527[   84.711852] binder: send failed reply for transaction 65520 to 1158:1495
08-23 20:18:45.620092[   84.730054] binder_alloc: 1158: binder_alloc_buf size 1056768 failed, no address space
08-23 20:18:45.620279[   84.739806] binder_alloc: allocated: 0 (num: 0 largest: 0), free: 1040384 (num: 1 largest: 1040384)
08-23 20:18:45.630325[   84.751296] binder: 1214:1489 transaction failed 29201/-28, size 1056768-0 line 2906
08-23 20:18:45.640247[   84.762580] binder: send failed reply for transaction 65598 to 1158:1495
08-23 20:18:45.868245[   84.986747] logd: logdr: UID=10086 GID=10086 PID=1214 n tail=0 logMask=4 pid=0 start=0ns
    deadline=0ns
08-23 20:18:46.004350[   85.125052] PM: suspend entry (deep)
08-23 20:18:46.095379[   85.214302] Filesystems sync: 0.083 seconds
08-23 20:18:46.095595[   85.219789] PM: Preparing system for sleep (deep)
08-23 20:18:54.728728[    2.149179] usbcore: registered new interface driver usbhid
08-23 20:18:54.729716[    2.156061] usbhid: USB HID core driver
08-23 20:18:54.729762[    2.161350] ashmem: initialized
08-23 20:18:54.729800[    2.167228] gnss: GNSS driver registered with major 235
08-23 20:18:54.729838[    2.174091] usbcore: registered new interface driver snd-usb-audio
08-23 20:18:54.729876[    2.182105] GACT probability NOT on
08-23 20:18:54.729909[    2.186429] Mirror/redirect action on
08-23 20:18:54.729942[    2.190839] netem: version 1.3
08-23 20:18:54.729979[    2.194749] u32 classifier
08-23 20:18:54.730012[    2.198488] input device check on
08-23 20:18:54.730044[    2.203101] Actions configured
08-23 20:18:54.730077[    2.208423] xt_time: kernel timezone is -0000
08-23 20:18:54.730114[    2.214123] ipip: IPv4 and MPLS over IPv4 tunneling driver

08-23 20:18:54.860930[   86.293218] [XR_BT_LPM] bluesleep_suspend:
08-23 20:18:54.860975[   86.300928] PM: suspend of devices complete after 1039.040 msecs
08-23 20:18:54.861027[   86.308509] PM: start suspend of devices complete after 1047.553 msecs
08-23 20:18:54.861074[   86.316667] sysrq: Terminate All Tasks
08-23 20:18:54.861112[   86.318318] PM: late suspend of devices complete after 1.656 msecs
08-23 20:18:54.861150[   86.334149] PM: noirq suspend of devices complete after 1.761 msecs
08-23 20:18:54.861188[   86.342010] Disabling non-boot CPUs ...

uncrypt.rc文件：
```

```
service uncrypt /system/bin/uncrypt
  class main
  socket uncrypt stream 600 system system
  disabled
  oneshot

vndservicemanager.rc文件：
service vndservicemanager /vendor/bin/vndservicemanager /dev/vndbinder
  class core
  user system
  group system readproc
  task_profiles ServiceCapacityLow
  onrestart class_restart main
  onrestart class_restart hal
  onrestart class_restart early_hal
  shutdown critical
```

5.1.4 问题解决方案

```
触发sysrq这个动作，是由一个debug补丁引入的，把该补丁去掉就不会复现了。

串口开启sysrq功能，休眠唤醒的时候，会触发sysrq，当往sysrq写e的时候，就会杀死所有的进程。
lichee/linux-5.4 / drivers/tty/serial/sunxi-uart.c
#if IS_ENABLED(CONFIG_SERIAL_SUNXI_CONSOLE) && IS_ENABLED(CONFIG_MAGIC_SYSRQ)
#define SUPPORT_SYSRQ
#endif

#include <linux/kernel.h>
#include <linux/module.h>
```

5.2 点击恢复出厂设置之后机器一直黑屏并且没有重启到 recovery 界面

5.2.1 问题描述

```
Settings应用中点击恢复出厂设置之后，机器一直黑屏，没有重启到recovery界面。
```

5.2.2 问题初步分析

```
从串口log看，机器并没有挂，按一下电源键，机器有被唤醒起来，但是过一会又被休眠下去了。
由于没有抓到前面的串口log，无法知道问题的原因，所以机器接上串口，抓取串口log进一步复测。

[ 368.155054][ T5223] [STA_WRN] xradio_flush vif is null return
[ 368.155246][ T5169] enable_sensor: sc7a20_CTRL_PWRDN
[ 368.155688][ T5169] hci: ERR: hcd is null
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

```
[ 368.155692][ T5169] hci: ERR: hcd is null
[ 368.155698][ T5169] udc is disable, need not enter to suspend
[ 368.155703][ T5169] G2D: g2d_suspend.
[ 368.155705][ T5169] G2D: g2d_suspend succesfully.
[ 368.155710][ T5169] [cedar] standby suspend
[ 368.155791][  T472] disp_suspend
[ 368.277322][  T472] disp_suspend finish
[ 368.808828][  T158] gslX680 suspend finished
[ 368.809118][ T5169] [NAND][NAND] nand_suspend
[ 368.809122][ T5169] [ND]RawNandHwSuperStandby start
[ 368.809137][ T5169] [ND]mode 2 : ONFI DDR/DDR2 => Async
[ 368.809278][ T5169] nand disable voltage vcc-nand
[ 368.809283][ T5169] nand disable voltage vcc-io
[ 368.809287][ T5169] nand had already disable voltage
[ 368.809288][ T5169] [NAND][NAND] nand_suspend ok
[ 368.809974][ T5169] sw_uart_suspend()2176 - uart1 suspend
[ 368.809989][ T5169] sw_uart_suspend()2176 - uart0 suspend
[ 368.839384][ T5169] NSI_PMU 3100000.nsi-controller: suspend okay
[ 368.840151][ T5169] [XR_BT_LPM] bluesleep_suspend:
[ 368.844310][ T5169] Disabling non-boot CPUs ...
[ 368.845037][    C2] CPU3: update max cpu_capacity 1024
[ 368.845216][ T5169] CPU1 killed.
[ 368.846935][ T5169] CPU2 killed.
[ 368.848638][ T5169] CPU3 killed.
[ 368.849507][    C0] PM: pm_system_irq_wakeup: 364 triggered axp803_irq_chip
[ 368.849556][ T5169] Enabling non-boot CPUs ...
[ 368.850436][ T5169] CPU1 is up
[ 368.851231][ T5169] CPU2 is up
[ 368.851919][ T5169] CPU3 is up
[ 368.852995][    C1] CPU1: update max cpu_capacity 1024
[ 368.854231][ T5169] [XR_BT_LPM] bluesleep_resume:
[ 368.854605][ T5169] NSI_PMU 3100000.nsi-controller: set limit to fit 720P(ABCDE)
[ 368.854609][ T5169] NSI_PMU 3100000.nsi-controller: resume okay
[ 368.856340][  T158] disp_resume finish
[ 368.856392][ T5222] disp_al_manager_apply ouput_type:0
[ 368.856577][ T5222] [DISP] lcd_clk_config,line:736:
[ 368.856584][ T5222] disp 0, clk: pll(408000000),clk(408000000),dclk(68000000) dsi_rate(68000000)
[ 368.856584][ T5222]      clk real:pll(408000000),clk(408000000),dclk(102000000) dsi_rate(150000000)
[ 368.893002][ T5169] sw_uart_resume()2203 - uart0 resume. DLH: 0, DLL: 13.
[ 368.893014][ T5169] sw_uart_resume()2203 - uart1 resume. DLH: 0, DLL: 0.
[ 368.893105][ T5169] sunxi-mmc 4021000.sdmmc: dat3_imask 0
[ 368.893117][ T5169] [NAND][NAND] nand_resume
[ 368.893120][ T5169] [ND]RawNandHwSuperResume start
[ 368.893128][ T5169] nand:enable voltage vcc-nand ok:a4e19dc6
[ 368.893132][ T5169] nand:enable voltage vcc-io ok:9f38d17f
[ 368.893134][ T5169] nand:has already enable voltage
[ 368.893198][ T5169] nand_clk_request
[ 368.893208][ T5169] nand_clk_request: get pll rate 1200000000HZ
[ 368.893255][ T5169] [ND]Reset NDFC start 0  0
[ 368.893259][ T5169] [ND]Reset NDFC end 0    0
[ 368.894477][ T5169] [ND]build nand physical connection ok...
[ 368.894478][ T5169] [ND]CE:
[ 368.894480][ T5169] [ND]0
[ 368.894481][ T5169] [ND]
[ 368.894481][ T5169] RB:
[ 368.894483][ T5169] [ND]0
[ 368.894485][ T5169] [ND]
[ 368.896698][ T5169] [ND]check scan data, first_check 4 0!
[ 368.896703][ T5169] [ND]336 interface change ddr_type:2 sclk0:75 sclk1:150 sdr_edo_bak:1 ddr_edo_bak:1
    ddr_delay_bak:0 !
```

```
[ 368.896707][ T5169] [ND]mode 1 : Async => ONFI DDR/DDR2
[ 368.896709][ T5169] [ND]mode 1 : Async => ONFI DDR/DDR2
[ 368.898046][ T5169] micron retry!
[ 368.898064][ T5169] set feature(addr 137) p0: 0 readretry Configuration failed!
[ 368.898451][ T5169] set feature(addr 137) p0: 0 readretry Configuration failed!
[ 368.898838][ T5169] set feature(addr 137) p0: 2 readretry Configuration failed!
[ 368.899225][ T5169] set feature(addr 137) p0: 4 readretry Configuration failed!
[ 368.899754][ T5169] set feature(addr 137) p0: 6 readretry Configuration failed!
[ 368.900335][ T5169] set feature(addr 137) p0: 8 readretry Configuration failed!
[ 368.900910][ T5169] set feature(addr 137) p0: 0 readretry Configuration failed!
[ 368.900915][ T5169] [NE]ecc err!read page, read page end error -2,chip=0 block=4 page=0
[ 368.900918][ T5169] [ND]ddr edo:0x4, delay:0x2 is wrong:-2
[ 368.901269][ T5169] [ND]ddr edo:0x4, delay:0x4 is good
[ 368.901573][ T5169] [ND]ddr edo:0x4, delay:0x6 is good
[ 368.901918][ T5169] [ND]ddr edo:0x4, delay:0x8 is good
[ 368.902208][ T5169] [ND]ddr edo:0x4, delay:0xa is good
[ 368.902498][ T5169] [ND]ddr edo:0x4, delay:0xc is good
[ 368.902787][ T5169] [ND]ddr edo:0x4, delay:0xe is good
[ 368.903077][ T5169] [ND]ddr edo:0x4, delay:0x10 is good
[ 368.903367][ T5169] [ND]ddr edo:0x4, delay:0x12 is good
[ 368.903656][ T5169] [ND]ddr edo:0x4, delay:0x14 is good
[ 368.903946][ T5169] [ND]ddr edo:0x4, delay:0x16 is good
[ 368.904235][ T5169] [ND]ddr edo:0x4, delay:0x18 is good
[ 368.904525][ T5169] [ND]ddr edo:0x4, delay:0x1a is good
[ 368.904842][ T5169] [ND]read page ecc limit,chip=0 block=4 page=0
[ 368.904844][ T5169] [ND]ddr edo:0x4, delay:0x1c is good
[ 368.905210][ T5169] micron retry!
[ 368.910381][ T5169] [NE]ecc err!read page, read page end error -2,chip=0 block=4 page=0
[ 368.910384][ T5169] [ND]ddr edo:0x4, delay:0x1e is wrong:-2
[ 368.910749][ T5169] micron retry!
[ 368.915917][ T5169] [NE]ecc err!read page, read page end error -2,chip=0 block=4 page=0
[ 368.915920][ T5169] [ND]ddr edo:0x4, delay:0x20 is wrong:-2
[ 368.916285][ T5169] micron retry!
[ 368.916301][ T5169] set feature(addr 137) p0: 0 readretry Configuration failed!
[ 368.916317][ T5169] set feature(addr 137) p0: 0 readretry Configuration failed!
[ 368.916334][ T5169] set feature(addr 137) p0: 0 readretry Configuration failed!
[ 368.916350][ T5169] set feature(addr 137) p0: 1 readretry Configuration failed!
[ 368.916367][ T5169] set feature(addr 137) p0: 2 readretry Configuration failed!
[ 368.916383][ T5169] set feature(addr 137) p0: 3 readretry Configuration failed!
[ 368.916399][ T5169] set feature(addr 137) p0: 4 readretry Configuration failed!
[ 368.916416][ T5169] set feature(addr 137) p0: 5 readretry Configuration failed!
[ 368.916432][ T5169] set feature(addr 137) p0: 6 readretry Configuration failed!
[ 368.916448][ T5169] set feature(addr 137) p0: 7 readretry Configuration failed!
[ 368.916465][ T5169] set feature(addr 137) p0: 8 readretry Configuration failed!
[ 368.916481][ T5169] set feature(addr 137) p0: 12 readretry Configuration failed!
[ 368.916498][ T5169] set feature(addr 137) p0: 0 readretry Configuration failed!
[ 368.916502][ T5169] [NE]ecc err!read page, read page end error -72,chip=0 block=4 page=0
[ 368.916504][ T5169] [ND]ddr edo:0x4, delay:0x22 is wrong:-72
[ 368.916594][ T5169] [ND](0x4, 0x4 - 0x1c)
[ 368.916597][ T5169] [ND]1738 good edo: 0x4, good delay chain: 0x11
[ 368.916602][ T5169] [ND]RawNandHwSuperResume end
[ 368.916604][ T5169] [NAND][NAND] nand_resume ok
[ 368.916738][ T5169] [cedar] standby resume
[ 368.916744][ T5169] G2D: g2d_resume.
[ 368.916746][ T5169] G2D: g2d_resume succesfully.
[ 368.916751][ T5169] udc is disable, need not enter to resume
[ 368.916755][ T5169] hci: ERR: hcd is null
[ 368.916762][ T5169] hci: ERR: hcd is null
[ 368.918638][ T5169] enable_sensor: sc7a20_CTRL_PWRDN
[ 368.918643][ T5247] gslX680 resume finished
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

```
[ 369.641304][ T5169] OOM killer enabled.
[ 369.650235][ T5169] Restarting tasks ...
[ 369.650332][  T131] sunxi-mmc 4020000.sdmmc: sdc set ios:clk 0Hz bm PP pm UP vdd 22 width 1 timing LEGACY(
    SDR12) dt B
[ 369.657690][ T5169] done.
[ 369.967677][ T5169] PM: suspend entry (deep)
[ 369.994011][ T5169] Filesystems sync: 0.021 seconds
[ 369.999721][ T5169] Freezing user space processes ... (elapsed 0.002 seconds) done.
[ 370.010435][ T5169] OOM killer disabled.
[ 370.014823][ T5169] Freezing remaining freezable tasks ... (elapsed 0.001 seconds) done.
[ 370.025511][ T5169] printk: Suspending console(s) (use no_console_suspend to debug)
```

5.2.3 问题排查步骤

机器接串口复现之后，从串口log可以看到，机器已经进入到恢复出厂设置的整个流程了，只是机器黑屏并进入休眠，导致机器没法进一步走到reboot流程。

也就是机器卡在了uncrypt阶段往misc分区写wipe-data数据，到reboot阶段之间，由于机器进入休眠，无法进一步执行。如果一直频繁按power键，系统是可以重启的。

机器进入到uncrypt服务阶段的log：

```
[ 384.480645][    T1] init: starting service 'setup-bcb'...
[ 384.501093][    T1] init: Created socket '/dev/socket/uncrypt', mode 600, user 1000, group 1000
[ 384.522933][    T1] init: Control message: Processed ctl.start for 'setup-bcb' from pid: 465 (system_server)
[ 385.599536][    T1] init: Service 'setup-bcb' (pid 5563) exited with status 0 oneshot service took 1.084000 seconds in
    background
[ 385.619476][    T1] init: Sending signal 9 to service 'setup-bcb' (pid 5563) process group...
```

系统休眠的log：

```
[ 393.035116][ T4192] [ND]RawNandHwSuperResume end
[ 393.035118][ T4192] [NAND][NAND] nand_resume ok
[ 393.035254][ T4192] [cedar] standby resume
[ 393.035260][ T4192] G2D: g2d_resume.
[ 393.035263][ T4192] G2D: g2d_resume succesfully.
[ 393.035268][ T4192] udc is disable, need not enter to resume
[ 393.035273][ T4192] hci: ERR: hcd is null
[ 393.035280][ T4192] hci: ERR: hcd is null
[ 393.037147][ T4192] enable_sensor: sc7a20_CTRL_PWRDN
[ 393.037154][    T7] gslX680 resume finished
[ 393.666100][ T4192] OOM killer enabled.
[ 393.675087][ T4192] Restarting tasks ...
[ 3
    UUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUUURUUUUUUUUUUUUUUUUUUU
```

```
[ 395.456751][ T4192] printk: Suspending console(s) (use no_console_suspend to debug)
```

5.2.4 问题解决方案

```
进入关机流程前，先把屏幕关掉，导致机器有可能因为休眠而无法关机。

解决办法：方案中配置，关机过程中，不黑屏

diff --git a/ad86310vat/overlay/overlay/frameworks/base/core/res/res/values/config.xml b/ad86310vat/overlay/overlay/
        frameworks/base/core/res/res/values/config.xml
index dc01e83..03785e8 100644
--- a/ad86310vat/overlay/overlay/frameworks/base/core/res/res/values/config.xml
+++ b/ad86310vat/overlay/overlay/frameworks/base/core/res/res/values/config.xml
@@ -727,7 +727,7 @@
      <item>255</item>
    </array>

-   <bool name="config_shutdownForceScreenOff">true</bool>
+   <bool name="config_shutdownForceScreenOff">false</bool>

    <string-array name="config_packagesExemptFromSuspension" translatable="false">
      <item>com.google.android.apps.wellbeing</item>
diff --git a/ad86310vc/overlay/overlay/frameworks/base/core/res/res/values/config.xml b/ad86310vc/overlay/overlay/
        frameworks/base/core/res/res/values/config.xml
index fe9284f..654a4e6 100644
--- a/ad86310vc/overlay/overlay/frameworks/base/core/res/res/values/config.xml
+++ b/ad86310vc/overlay/overlay/frameworks/base/core/res/res/values/config.xml
@@ -727,7 +727,7 @@
      <item>255</item>
    </array>

-   <bool name="config_shutdownForceScreenOff">true</bool>
+   <bool name="config_shutdownForceScreenOff">false</bool>

    <string-array name="config_packagesExemptFromSuspension" translatable="false">
      <item>com.google.android.apps.wellbeing</item>
diff --git a/common/overlay/overlay/frameworks/base/core/res/res/values/config.xml b/common/overlay/overlay/
        frameworks/base/core/res/res/values/config.xml
index 78893b4..37db6d3 100755
--- a/common/overlay/overlay/frameworks/base/core/res/res/values/config.xml
+++ b/common/overlay/overlay/frameworks/base/core/res/res/values/config.xml
@@ -728,7 +728,7 @@
      <item>255</item>
    </array>

-   <bool name="config_shutdownForceScreenOff">true</bool>
+   <bool name="config_shutdownForceScreenOff">false</bool>

    <string-array name="config_packagesExemptFromSuspension" translatable="false">
      <item>com.google.android.apps.wellbeing</item>

代码流程：
frameworks/base/services/core/java/com/android/server/power/ShutdownThread.java
      /// AW CODE:[feat]:shut down screen directly
      final boolean shutdownForceScreenOff = context.getResources().getBoolean(
            com.android.internal.R.bool.config_shutdownForceScreenOff);
```

```
    final boolean showShutdownAnimation = SystemProperties.getBoolean("ro.init.shutdown_animation", false);
    if (shutdownForceScreenOff && !showShutdownAnimation) {
        sInstance.mPowerManager.goToSleep(SystemClock.uptimeMillis(),
            PowerManager.GO_TO_SLEEP_REASON_POWER_BUTTON, 0);
    }
```

5.3 恢复出厂设置失败

5.3.1 问题描述

恢复出厂设置失败了

5.3.2 问题初步分析

从串口log看，60多秒还没跑完，应该是卡在了格式化的流程中。

```
[   7.235222][  T155] EXT4-fs (mmcblk0p19): mounted filesystem with ordered data mode. Opts: errors=remount-ro,
        nomblk_io_submit
[   7.235599][  T155] ext4 filesystem being mounted at /metadata supports timestamps until 2038 (0x7fffffff)
[   7.241871][  T155] EXT4-fs (mmcblk0p19): mounted filesystem with ordered data mode. Opts: errors=panic
[   7.242200][  T155] ext4 filesystem being mounted at /metadata supports timestamps until 2038 (0x7fffffff)
[  11.018533][   T1] init: starting service 'multi_ir'...
[  11.026304][  T178] init: cpuset cgroup controller is not mounted!
[  11.033665][   T32] kauditd_printk_skb: 1142 callbacks suppressed
[  11.033672][   T32] audit: type=1400 audit(1666056680.164:80): avc:  denied  { entrypoint } for  pid=178 comm="init"
        path="/system/bin/multi_ir" dev="rootfs" ino=2162 scontext=u:r:multi_ir:s0 tcontext=u:object_r:rootfs:s0 tclass=
        file permissive=1
[  11.043249][  T178] input: sunxi-ir-uinput as /devices/virtual/input/input1
[  33.943673][   T55] usb0-drvvbus: disabling
[  57.845363][    C0] random: crng init done
[  57.849969][    C0] random: 7 urandom warning(s) missed due to ratelimiting
```

5.3.3 问题排查步骤

1. 进入recovery之后，接adb，然后执行adb root
2. 输入ps -Af | grep make_f2fs
3. cat /proc/make_f2fs_pid/stack 可以看到make_f2fs卡在了discard的ioctl中

所以有可能是这块emmc不支持BLKSECDISCARD ioctl

版权所有 © 珠海全志科技股份有限公司。保留一切权利

5.3.3.1 问题解决方案

```
解决办法：
把恢复出厂设置中，可能调用到BLKSECDISCARD ioctl的地方都去掉，这样就不会卡住了。

android/external/f2fs-tools目录：
diff --git a/mkfs/f2fs_format_utils.c b/mkfs/f2fs_format_utils.c
index f2d55ad..7e69618
--- a/mkfs/f2fs_format_utils.c
+++ b/mkfs/f2fs_format_utils.c
@@ -86,7 +86,7 @@ static int trim_device(int i)
                free(stat_buf);
                return f2fs_reset_zones(i);
        }
-#ifdef BLKSECDISCARD
+#if 0
        if (ioctl(fd, BLKSECDISCARD, &range) < 0) {
                MSG(0, "Info: This device doesn't support BLKSECDISCARD\n");
        } else {

services/core/jni/com_android_server_PersistentDataBlockService.cpp
diff --git a/services/core/jni/com_android_server_PersistentDataBlockService.cpp b/services/core/jni/
    com_android_server_PersistentDataBlockService.cpp
index 97e69fb4bb40..d44f5997d29e 100644
--- a/services/core/jni/com_android_server_PersistentDataBlockService.cpp
+++ b/services/core/jni/com_android_server_PersistentDataBlockService.cpp
@@ -57,12 +57,12 @@ namespace android {
    if (range[1] == 0)
      return 0;

-     ret = ioctl(fd, BLKSECDISCARD, &range);
+     ret = ioctl(fd, BLKGETSIZE64, &range);
    if (ret < 0) {
      ALOGE("Something went wrong secure discarding block: %s\n", strerror(errno));
      range[0] = 0;
      range[1] = len;
-     ret = ioctl(fd, BLKDISCARD, &range);
+     ret = ioctl(fd, BLKGETSIZE64, &range);
      if (ret < 0) {
        ALOGE("Discard failed: %s\n", strerror(errno));
        return -1;
```

5.4 刷完谷歌 gsi 后，无法在设置界面恢复出厂设置

5.4.1 问题描述

```
1.平板正常烧录固件开机
2.通过fastboot方式烧入 Google system.img和boot-5.4.img
3,烧入完成后开机进入系统，修改待机时间，锁屏方式，桌面布局等
4.点击 设置-系统-重置选项-清除所有数据(恢复出厂设置)，机器进入重启
5.等机器重启发现之前修改的设置项和桌面布局等修改都未还原，恢复出厂设置功能失败
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

5.4.2 问题初步分析

从串口log看，系统并没有进入到recovery模式，而是直接正常进系统了。

因为Google提供的GKI（也就是boot-5.4.img）没有/system/bin/recovery，所以没办法进入到recovery，执行恢复出厂设置。
烧了GKI之后，如果想恢复出厂设置，可以执行：
$ reboot bootloader
$ fastboot -w
$ fastboot reboot

解压boot-5.4.img可以看到Google提供的GKI是没有recovery程序的：
source build/envsetup.sh 并 lunch之后执行：
$ mkdir out_boot
$ unpack_bootimg --boot_img boot-5.4.img --out out_boot --format=mkbootimg -0
$ cd out_boot
$ lz4 -dc ramdisk | cpio -i
$ find -name recovery 是找不到recovery的

对于我们自己编译出来的boot.img（如out/target/product/ceres-p25/目录下），执行上面的命令，是有recovery程序的，所以可以进入到recovery，执行恢复出厂设置并恢复出厂成功

5.5 开机过程中卡在了电池图标界面

5.5.1 问题描述

开机过程中，卡在了电池图标界面

5.5.2 问题初步分析

进recovery的同时进charger就会卡在电池图标界面。

cmdline进recovery和进charger的标志：
recovery的标志：androidboot.force_normal_boot=0
charger的标志：androidboot.mode=charger bootreason=charger

5.5.3 问题排查步骤

手动复现现场：
（1）执行reboot recovery
（2）按s进入uboot控制台
（3）输入setenv cma 8M androidboot.mode=charger bootreason=charger androidboot.selinux=permissive
（4）saveenv 并 boot
（5）不会执行recovery并进入到recovery界面

版权所有 © 珠海全志科技股份有限公司。保留一切权利

总结：恢复出厂设置 -> 重启进入recovery -> uboot阶段同时判断进入充电模式了。（recovery + charger）就会出现这种问题。

5.5.4 问题解决方案

解决办法：不是recovery模式才进charger

```
diff --git a/init/init.cpp b/init/init.cpp
old mode 100644
new mode 100755
index f552994..0a9ec02
--- a/init/init.cpp
+++ b/init/init.cpp
@@ -897,7 +897,7 @@ int SecondStageMain(int argc, char** argv) {

    // Don't mount filesystems or start core system services in charger mode.
    std::string bootmode = GetProperty("ro.bootmode", "");
-   if (bootmode == "charger") {
+   if (!IsRecoveryMode() && bootmode == "charger") {
        am.QueueEventTrigger("charger");
    } else {
        am.QueueEventTrigger("late-init");
```

广州星翼电子科技有限公司zdyzlinux

6 OTA 问题

6.1 ota 差分包升级异常

6.1.1 问题描述

删除APK，OTA升级失败。

log报错显示：

```
[ 12.791178] creating stash /cache/recovery/cc46ebfd04058569d0c6c1431c6af6c1328458e4
[ 12.792286] 819957760 bytes free on /cache (92819456 needed)
[ 12.792563] /cache/recovery/last_command doesn't exist.
[ 12.796038] moving 17 blocks
[ 12.796112] Failed to write 69632 bytes of data: Operation not permitted
[ 12.796123] failed to execute command [move b293ce8c9b8e0cbf1e87fb6e119682af3c7fdbca 2,34546,34563 17
        2,39904,39921]
[ 12.796132] new data receiver is still available after executing all commands.
[ 12.798700] script aborted: E1001: Failed to update system image.
[ 12.860570] E:Error in /cache/update.zip (status 7)
[ 12.893395]
[ 12.918345] W:fai
```

6.1.2 问题初步分析

从log看，recovery在执行move操作，往system分区写数据的时候，显示没有权限。只能从源码做进一步分析了。

```
[ 12.791178] creating stash /cache/recovery/cc46ebfd04058569d0c6c1431c6af6c1328458e4
[ 12.792286] 819957760 bytes free on /cache (92819456 needed)
[ 12.792563] /cache/recovery/last_command doesn't exist.
[ 12.796038] moving 17 blocks
[ 12.796112] Failed to write 69632 bytes of data: Operation not permitted
[ 12.796123] failed to execute command [move b293ce8c9b8e0cbf1e87fb6e119682af3c7fdbca 2,34546,34563 17
        2,39904,39921]
[ 12.796132] new data receiver is still available after executing all commands.
[ 12.798700] script aborted: E1001: Failed to update system image.
[ 12.860570] E:Error in /cache/update.zip (status 7)
[ 12.893395]
[ 12.918345] W:fai
```

版权所有 © 珠海全志科技股份有限公司。保留一切权利

6.1.3 问题排查步骤

1. 排查write系统调用为啥没有权限

（1）在安卓界面进行OTA升级的话，比较难调试，升级成功之后会自动重启。可以通过adb sideload命令来进行OTA升级，这样可以一直在recovery上进行操作，方便调试。

（2）开启内核ftrace功能，通过function_graph功能来排查write系统调用在内核的哪一个函数上返回错误码了。

经排查：
longan/kernel/linux-4.9/fs/block_dev.c

```
ssize_t blkdev_write_iter(struct kiocb *iocb, struct iov_iter *from)
{
   struct file *file = iocb->ki_filp;
   struct inode *bd_inode = bdev_file_inode(file);
   loff_t size = i_size_read(bd_inode);
   struct blk_plug plug;
   ssize_t ret;

// 在这里返回了permission deny
   if (bdev_read_only(I_BDEV(bd_inode)))
      return -EPERM;
```

longan/kernel/linux-4.9/block/genhd.c

```
int bdev_read_only(struct block_device *bdev)
{
   if (!bdev)
      return 0;
      // 这里设置为1了
   return bdev->bd_part->policy;
}
```

longan/kernel/linux-4.9/block/genhd.c

```
void set_disk_ro(struct gendisk *disk, int flag)
{
   struct disk_part_iter piter;
   struct hd_struct *part;

   if (disk->part0.policy != flag) {
      set_disk_ro_uevent(disk, flag);
      // 这里设置只读属性的
      disk->part0.policy = flag;
   }

   disk_part_iter_init(&piter, disk, DISK_PITER_INCL_EMPTY);
   while ((part = disk_part_iter_next(&piter)))
      part->policy = flag;
   disk_part_iter_exit(&piter);
}
```

通过在set_disk_ro加dump_stack，发现会通过ioctl来调用set_disk_ro的
longan/kernel/linux-4.9/drivers/md/md.c中的md_ioctl
所以就是安卓层设置dm设备的只读属性。

2. 排查安卓层设置dm设备为只读

（1）通过strace recovery进程，查看ioctl命令，发现会通过ioctl 设置dm设备的table为readonly的

```
strace -p recovery_pid -s 400 -tt -T -f -v
```

（2）查看代码，发现会设置DM_READONLY_FLAG只读flag
android/system/core/fs_mgr/libdm/dm.cpp的LoadTableAndActivate函数

```
  if (table.readonly()) {
    io->flags |= DM_READONLY_FLAG;
  }
```

android/system/core/fs_mgr/fs_mgr_dm_linear.cpp
CreateDmTable函数：
如果super分区设置了只读参数，一般super分区都会设置，所以默认就是只读的

```
  if (partition.attributes & LP_PARTITION_ATTR_READONLY) {
    table->set_readonly(true);
  }
```

CreateLogicalPartition函数：

```
  if (!CreateDmTable(metadata, partition, super_device, &table)) {
    return false;
  } // 如果force_writable参数为true，则会去掉只读模式
  if (force_writable) {
    table.set_readonly(false);
  }
```

（3）通过查看升级包里面的updater-script文件，map_partition函数会去挂载分区。
android/bootable/recovery/updater/dynamic_partitions.cpp
map_partition
  -> MapPartitionFn
    -> MapPartitionOnDeviceMapper
      -> CreateLogicalPartition ->设置force writable为true

（4）对比正常和异常的log
发现异常的log，会在Installing update...前进行分区的挂载，而正常的log，会在Installing update...后进行分区的挂载。

然后查看源码发现：
android/bootable/recovery/install/install.cpp
其会在verify_package_compatibility函数中进行system/vendor/product分区的挂载，并且force_writeable为false，也就是挂成只读了。

然后updater-script文件，map_partition函数去挂载分区的时候，判断分区已经挂载的话，就不会再次挂载了。

（5）排查verify_package_compatibility函数为啥会挂载vendor分区
verify_package_compatibility函数是解析升级包的compatibility.zip文件的，当vendor_manifest.xml文件不存在的时候，就会去挂载vendor分区。

（6）排查生成ota包的时候，什么时候会生成vendor_manifest.xml文件
android/build/make/tools/releasetools/ota_from_target_files.py
AddCompatibilityArchive函数中，可以看到，当vendor分区的fingerprint改变的时候，才会生成vendor_manifest.xml文件

6.1.4 问题解决方案

由于需要固定fingerprint，所以vendor或者system的fingerprint肯定是不会改变的。
这样只需要修改下ota_from_target_files.py，让它默认生成vendor_manifest.xml文件就好了。

```
diff --git a/tools/releasetools/ota_from_target_files.py b/tools/releasetools/ota_from_target_files.py
index 8b55f034b3..ab4ad5fe4d 100755
--- a/tools/releasetools/ota_from_target_files.py
+++ b/tools/releasetools/ota_from_target_files.py
@@ -831,12 +831,11 @@ def AddCompatibilityArchiveIfTrebleEnabled(target_zip, output_zip, target_info,
```

```
  source_fp = source_info.fingerprint
  target_fp = target_info.fingerprint
-  system_updated = source_fp != target_fp
+  system_updated = True

  # other build fingerprints could be possibly blacklisted at build time. For
  # such a case, we consider those images being changed.
-  vendor_updated = FingerprintChanged(source_info.vendor_fingerprint,
-                                      target_info.vendor_fingerprint)
+  vendor_updated = True
  product_updated = HasProductPartition(target_zip) and \
                    FingerprintChanged(source_info.product_fingerprint,
                                       target_info.product_fingerprint)
```

广州星翼电子科技有限公司zdyzlinux

著作权声明

版权所有 ©2023 珠海全志科技股份有限公司。保留一切权利。

本文档及内容受著作权法保护，其著作权由珠海全志科技股份有限公司（“全志”）拥有并保留一切权利。

本文档是全志的原创作品和版权财产，未经全志书面许可，任何单位和个人不得擅自摘抄、复制、修改、发表或传播本文档内容的部分或全部，且不得以任何形式传播。

商标声明

、、全志科技、（不完全列举）均为珠海全志科技股份有限公司的商标或者注册商标。在本文档描述的产品中出现的其它商标，产品名称，和服务名称，均由其各自所有人拥有。

免责声明

您购买的产品、服务或特性应受您与珠海全志科技股份有限公司（“全志”）之间签署的商业合同和条款的约束。本文档中描述的全部或部分产品、服务或特性可能不在您所购买或使用的范围内。使用前请认真阅读合同条款和相关说明，并严格遵循本文档的使用说明。您将自行承担任何不当使用行为（包括但不限于如超压，超频，超温使用）造成的不利后果，全志概不负责。

本文档作为使用指导仅供参考。由于产品版本升级或其他原因，本文档内容有可能修改，如有变更，恕不另行通知。全志尽全力在本文档中提供准确的信息，但并不确保内容完全没有错误，因使用本文档而发生损害（包括但不限于间接的、偶然的、特殊的损失）或发生侵犯第三方权利事件，全志概不负责。本文档中的所有陈述、信息和建议并不构成任何明示或暗示的保证或承诺。

本文档未以明示或暗示或其他方式授予全志的任何专利或知识产权。在您实施方案或使用产品的过程中，可能需要获得第三方的权利许可。请您自行向第三方权利人获取相关的许可。全志不承担也不代为支付任何关于获取第三方许可的许可费或版税（专利税）。全志不对您所使用的第三方许可技术做出任何保证、赔偿或承担其他义务。